

MEIKO AUTOMATION

Industrial IoT Warning System — Real-Time Audio & Visual Alerting
for Smart Factory Environments

MGSP-V1

MGLB-V1

MGWI-V1



Made with GAMMA

SYSTEM OVERVIEW

Built for the Industrial Edge

MEIKO AUTOMATION delivers a purpose-engineered IoT warning platform that connects AI detection systems to real-time audio and visual alerts — with a target latency of $\leq 1-2$ seconds.



AI Camera

AI Server

MQTT / HTTP

Audio & Visual

The architecture is designed for reliability-first deployment: PoE-powered, protocol-redundant, and remotely managed — ready for Smart Factory, CCTV-AI, and fire safety applications.

PRODUCT LINE

Three Devices. One Unified Platform.



MGSP-V1

IP Audio Alert Device

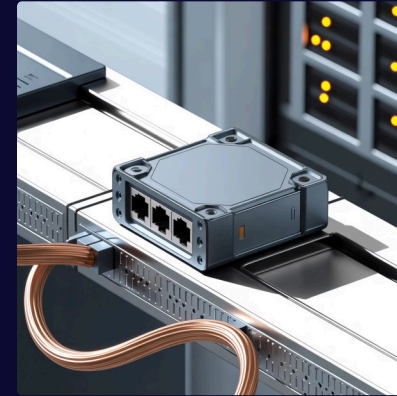
8W Class D speaker. MQTT + HTTP.
PoE powered. Offline fallback. ~90dB
@1m.



MGLB-V1

IP Visual Alert Device

RGB LED tower with buzzer.
ON/OFF/Blink/Pattern. PoE powered.
85–90dB buzzer.



MGWI-V1

IoT Alert Gateway

Ethernet + WiFi + RS485. MQTT
broker. Modbus integration. Full device
management.

PRODUCT 01

MGSP-V1 — IP Audio Alert Device

Industrial-grade IP speaker engineered for AI-triggered real-time audio notifications across factory floors and safety zones.

Key Capabilities

8W Class D Speaker

T-774W model · ~90dB
@1m · WAV/MP3 playback

Dual Protocol

MQTT + HTTP API · $\leq 1-2$
second latency

Resilient Design

Hardware watchdog ·
Offline audio fallback · ≥ 5
onboard files

Power & Connectivity

PoE 802.3af/at primary input

12V DC backup input

Ethernet RJ45 · 10/100 Mbps

WiFi (pending confirmation)

OTA firmware update supported

Over-voltage / under-voltage /
over-current / reverse polarity
protection

Dimensions: 120 × 88 × 38 mm

MGLB-V1 — IP Visual Alert Device

Industrial IP beacon with integrated buzzer — delivering unmistakable tri-color visual warnings synchronized in real time with AI detection events.

Alert Modes & Signal

 **Red**

Critical alarm /
Emergency

 **Yellow**

Warning / Caution zone

 **Green**

Normal / System OK

Modes: ON · OFF · Blink · Pattern · Buzzer 2W · 85–90dB @1m

Specifications

Beacon Model	ST45LF-BZ-3-24-RAG-LB18
Power Input	PoE 802.3af/at + 12V Backup
Consumption	25W System
Protocol	HTTP API / MQTT
Firmware	OTA Supported
Monitoring	Online / Offline
Dimensions	120 × 88 × 38 mm

PRODUCT 03

MGWI-V1 — IoT Alert Gateway

The intelligence hub of the MEIKO system — bridging AI servers, MQTT brokers, Modbus field devices, and IP alert endpoints in a single ruggedized unit.

Connectivity Matrix



Ethernet + WiFi

RJ45 10/100 · WiFi dual-mode



Modbus RS485

Field device integration



MQTT Broker

AI server ↔ device messaging

Protection Thresholds

Over-voltage protection: **27V**

Under-voltage protection: **7V**

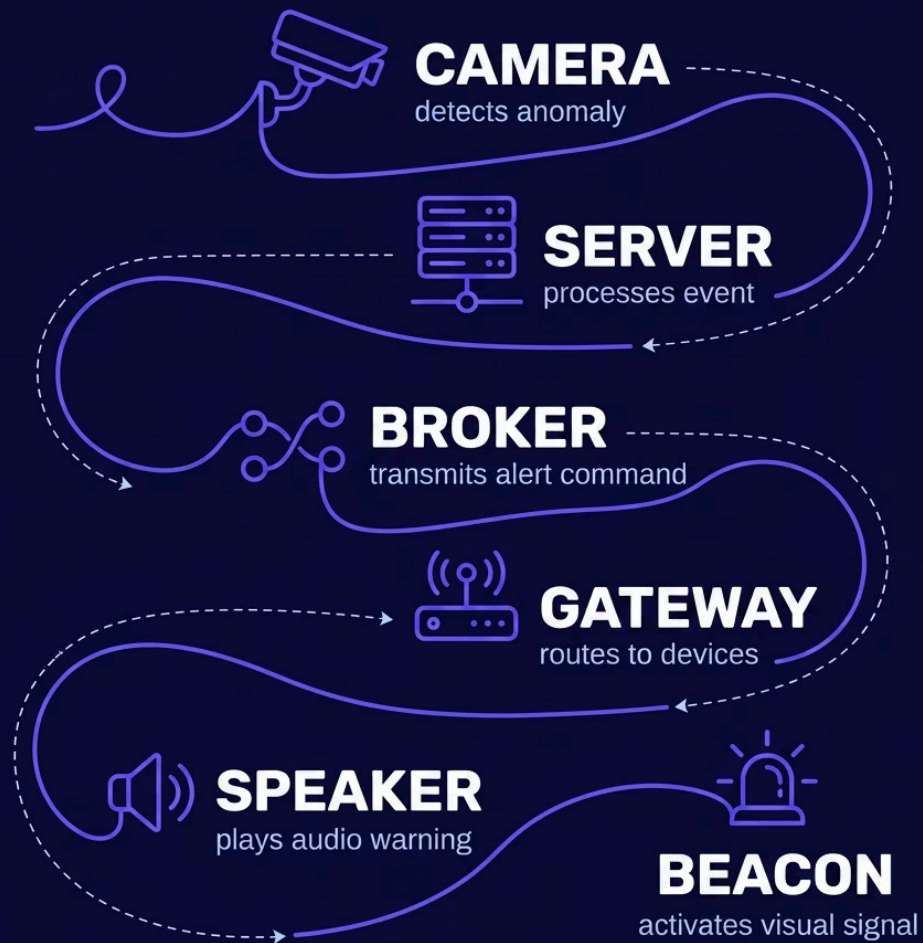
Over-current protection: **3.8A**

Reverse polarity: **Supported**

Power: PoE 802.3af/at + 12V backup · 25W

OTA firmware · Remote config · Audio + Beacon control

End-to-End AI Alert Workflow



Architecture Highlights

→ $\leq 1-2$ Second Latency

From AI detection event to audible/visual output — engineered for immediate response.

→ Protocol Redundancy

MQTT and HTTP API operate in parallel — no single point of communication failure.

→ Remote & Cloud Ready

Device health monitoring, OTA updates, and remote configuration via cloud dashboard or mobile.

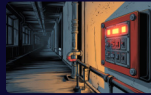
USE CASES

Deployed Across Critical Industrial Environments



Smart Factory & AI Camera Alert

CCTV-AI systems trigger instant audio and visual warnings on anomaly detection — no manual intervention required.



Fire & Safety Alarm Integration

Interfaces with fire detection panels and safety PLCs to issue multi-zone audio-visual evacuation alerts.



Restricted Area & Machine Safety

Real-time perimeter breach notification and machine safeguard alerts in hazardous zones and warehouses.

Comparative Specification Overview

Parameter	MGSP-V1	MGLB-V1	MGWI-V1
Device Type	IP Audio Alert	IP Visual Alert	IoT Gateway
Power Input	PoE 802.3af/at + 12V	PoE 802.3af/at + 12V	PoE 802.3af/at + 12V
Consumption	—	25W	25W
Network	Ethernet RJ45	Ethernet RJ45	Ethernet + WiFi
Wireless	WiFi (TBC)	—	WiFi Supported
Protocol	HTTP / MQTT	HTTP / MQTT	MQTT / Modbus RS485
Audio Output	8W · ~90dB @1m	Buzzer 2W · 85–90dB	Audio Control
Visual Output	—	Red/Yellow/Green LED	Beacon Control
OTA Firmware	Yes	Yes	Yes
Dimensions	120×88×38 mm	120×88×38 mm	120×88×38 mm

 All three devices share a common form factor (120 × 88 × 38 mm) and are protected against over-voltage, under-voltage, over-current, and reverse polarity conditions.



OEM / ODM & CONTACT

Your Platform. Your Brand.

OEM / ODM Capabilities

Full system integration support available. Designed for Smart Factory Integrators, CCTV-AI Providers, Fire Alarm Contractors, and Industrial Automation Partners.

Contact MEIKO AUTOMATION

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Solutions: MGSP-V1 · MGLB-V1 · MGWI-V1

Services: OEM · ODM · System Integration

- ❑ Enquire about volume pricing, custom firmware development, and dedicated integration support for your deployment.



Custom
Firmware



Private
Label



HW
Modification



Bulk Manufacturing